



ALLIANCE
UNIVERSITY

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NAAC
GRADE A+
ACCREDITED UNIVERSITY

COURSE OF INDEPENDENT STUDY

ODD SEMESTER 2026

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1. ABOUT THE COURSE

Course Overview: The Independent Study Course is an opportunity for students to engage in a research-driven or innovation-oriented learning experience beyond the conventional classroom structure and earn credits. This course allows a student to work closely with a faculty mentor solve real world problems through solutions approach without compromising the learning outcomes. Rather than attending a conventional classroom teaching, the student will:

A.

Explore a faculty-guided research/project problem, individually or in a small group.

B.

Work across disciplinary boundaries, combining knowledge from multiple domains.

C.

Appear for 2nd term assessment through either a research paper (SCOPUS-indexed conference or journal) with the help of CFR or a patent application with a TRL level of 5 or above (filed with the Indian Patent Office through the AU-IPR Cell).

Course Objectives

- > To promote a research mindset and nurture original thinking among undergraduate students.
- > To provide exposure to advanced topics and problem-solving in real-world contexts.
- > To develop self-learning, academic writing, project management, and communication skills.
- > To foster a culture of responsible research by encouraging attention to societal, environmental, and ethical aspects.

What Will Students Gain?

By enrolling in this course, a student can:

- > Replace traditional classroom learning with a customized, high-impact research experience.
- > Learn to conduct literature surveys, define problem scopes, design methods, validate results, and publish outcomes, skills highly valued in academia and industry.
- > Build a strong profile for higher studies, research internships, or entrepreneurship.
- > Get one-on-one mentorship from a faculty expert.
- > Graduate with a publication or patent, a significant addition to any resume or graduate school application.

2. ELIGIBILITY FOR THE COURSE

Students have the opportunity to undergo such courses through independent study mode as may be determined by the university from time to time.

Who Can Apply?

- > The University before the commencement of the semester shall provide the eligibility criteria for such courses.
- > Only students who are genuinely interested in guided research or problem-solving in a structured, goal-oriented environment are advised to apply.

Additional Requirements

A.

Students must meet the prerequisites (if any) set by the faculty who have floated the research problems

Prerequisites may include knowledge of specific subjects, software/tools, prior coursework, academic credentials, or experience

B.

Students must submit an Expression of Interest and undergo an interview conducted by the supervising faculty

C.

The selection is entirely at the discretion of the supervisor, who may

Choose not to select any candidate, or Assign the project to an individual student or a team of up to three students

3. PROCESS AND TIMELINES

This course follows a structured process, from problem selection to submission and evaluation. Students must adhere to each step as outlined below.

A. Announcement of Problems

- > At the beginning of the semester, a curated list of research problems or project topics will be released.

- > Each entry will include:

- 1 Title and description of the problem statement.
- 2 Name of the faculty supervisor who floated the problem.
- 3 Clearly defined prerequisites (skills, prior coursework, tools, etc.) required to undertake the research/project.

B. Expression of Interest (Eol) Submission

- > Students interested in a problem must submit an Expression of Interest (Eol) using the official format (to be shared)
 - > This submission is not a guarantee of selection — it simply conveys the student's interest and intent.
- > The Eol must include
 - 1 The selected problem statements.
 - 2 A brief justification for interest.
 - 3 Any prior experience or skills relevant to the problem.

C. Interview and Supervisor Selection

- > Faculty supervisors will interview applicants to assess
 - 1 Their motivation, background, and readiness for independent work.
 - 2 Their ability to meet the problem's technical or conceptual prerequisites.
- > Supervisors may
 - 1 Select or reject any applicant.
 - 2 Assign the problem to: A single student, or A team of 2–3 students (based on problem scope and dynamics)
 - > Final decision rests with the supervisor.

D. Probation Period (First 10 Days)

- > Selected students will be enrolled in the Independent research course on a 10-day
 - > During this period, the supervisor will evaluate:
 - 1 The student's engagement, independent effort, and interest.
 - 2 Adherence to instructions and quality of initial progress (e.g., literature review start).
- > At the end of 10 days:
 - 1 The supervisor may confirm continuation, or the student must then join the regular mode of the course.

E. Supervision and Log Maintenance

- > Students who continue after probation must
 - 1 Meet with their supervisor at least 3 times/week throughout the semester.
 - 2 Maintain a diary with: Date and time of meeting, Summary of discussion, Next steps and action points, Signature of the supervisor after every meeting.
- > Faculty will be available outside of class hours; students must schedule the meetings respectfully.

F. Milestone-Based Progress Assessment: The course is designed over a 13-week structure, divided into five major milestones. Each milestone will be evaluated using a standard evaluation rubric (already shared).teaching, the student will:

Milestone	Timeline
Literature Survey	Weeks 1–2
Scope/Objective/Problem Finalisation	Weeks 3–4
Methodology, Experimental Setup & Results	Weeks 5–8
Validation and Testing	Weeks 9–10
Paper Writing and Documentation	Weeks 10–13

A.

Review and evaluations will occur after each phase.

B.

Rubric-based feedback will be shared with the student.

G. Final Deliverable and Submission Requirement

- > Students must submit one of the following by the end of the 13th week
- 1 A research paper submitted to the Supervisor who will eventually communicate it to aSCOPUS indexed journal or conference, or A patent application filed with the Indian Patent Office through AU-IPR cell (with a minimum TRL level of 5)

- > Only upon successful submission of the research paper/ patent the student be allowed to appear for the End-Term Viva/Defence.
- > Failure to submit by the deadline will result in:
 - 1 Ineligibility for Viva
 - 2 The course being marked as Incomplete (NC) with allotment of a Fail grade ("F").

4. CONDITIONS FOR ELIGIBILITY

To be officially enrolled in the Independent Study Course, the student must fulfil the following conditions:

- > Meet the prerequisites (as defined in the problem listing) for the chosen problem.
- > Submit an Expression of Interest (EoI) and undergo an interview with the faculty supervisor.
- > Be selected by a faculty supervisor for a floated research/project problem.
- > Successfully complete the 10-day probation period, demonstrating seriousness, initiative, and potential to complete the course objectives.

5. CONDITIONS TO COMPLETE THE COURSE

To successfully complete the Independent Study Course and earn credit, students must:

- > Remain active and committed throughout the semester, adhering to the agreed plan.
- > Complete all five milestones within the designated timelines.
- > Attend at least 3 formal review meetings every week with the supervisor and maintain a logbook of meeting minutes
- > Submit a research output in one of the following forms

- 1** A research paper submission to SCOPUS through the Supervisor.
 - 2** A filed patent application (only for ideas with TRL level 5 or above) through AU-IPR Cell
- > Successfully appear for and clear the End-Term Viva/Defence based on the final output.
- > Comply with all academic and ethical standards defined by the University.

Note: Failure to submit the final output by Week 13 will automatically lead to the course being marked as Incomplete (NC), making the student ineligible for viva.

6. PROVISIONS FOR REAPPEARING IN THE COURSE

The provision for reappearing or re-registering for the Independent Study Course

- > Any student who receives a "Fail" grade "F" will be required to pay the examination fee and may appear for the viva only after fulfilling the eligibility criteria (i.e., paper submission).
- > Students who withdraws mid of the semester will be marked as "I" grade and has to repeat the open elective course in summer term.

7. ETHICAL CODE OF CONDUCT

Students enrolled in the Independent Study Course are expected to uphold the highest standards of academic and research ethics throughout the duration of their project.

Salient Requirements

- > Students must follow all ethical protocols and obtain necessary approvals from the Research Ethics committee (if required), as suggested, and guided by the supervisor.
- > This applies especially to projects involving

- 1** Human or animal subjects
 - 2** Sensitive data
 - 3** Third-party intellectual property
 - 4** Societal or environmental impact studies

Any deviation from the approved procedures without prior consent may result in:

A.

Withdrawal from the course

B.

Denial of submission or viva

C.

Institutional disciplinary action, as applicable

8. UNIVERSITY PLAGIARISM POLICY

All written submissions — including literature reviews, reports, final documentation, and research papers — must comply with the University's Academic Integrity and Plagiarism Policy.

Salient Requirements

- > Submissions must demonstrate original work and proper citation of all sources
- > Permissible similarity index
 - 1** 15% for interim submissions
 - 2** 10% for final paper or patent documentation
- > The use of generative AI tools (e.g., ChatGPT, Copilot) must be explicitly disclosed and used only with supervisor approval.
- > Plagiarism includes
 - 1** Copying without citation
 - 2** Rephrasing without attribution
 - 3** Submitting work not authored by the student/team
- > Violations may result in:
 - 1** Rejection of submission
 - 2** Ineligibility for End-Term Viva
 - Academic penalties or disciplinary proceedings

For complete details, refer to the University's Plagiarism Policy available with the Centre for Research or on the official university website.

MILESTONE: 1. LITERATURE SURVEY

Marks: 10

Duration: Week 1 and 2

Parameter: 1. Relevance of Sources (How well the sources relate to the selected problem)**DEVELOPING (1)**

The sources selected are either very few, outdated, or not meaningfully connected to the topic. They do not demonstrate adequate understanding of the field.

PROFICIENT (2)

The sources are mostly aligned with the selected topic and objectives. There is a reasonable effort to find relevant academic references, though some may be weak or tangential.

EXEMPLARY (2.5)

The sources are highly relevant, current, and directly support the understanding of the selected problem. They demonstrate clear familiarity with the core domain.

Parameter: 2. Diversity and Type of Sources (How varied and scholarly the sources are)**DEVELOPING (1)**

Most sources are of one type (e.g., websites, blogs) or from non-scholarly sources. Key academic references like journal articles, standards, or patents are missing.

PROFICIENT (2)

A reasonable mix is present, including some scholarly papers, reports, and academic articles. May lack depth or miss important source types.

EXEMPLARY (2.5)

A rich variety of credible sources is included — such as peer-reviewed journals, patents, technical standards, books, and research databases — indicating a comprehensive search.

MILESTONE: 1. LITERATURE SURVEY**Marks: 10****Duration:** Week 1 and 2**Parameter: 3. Thematic Organization** (How well the literature is grouped and structured)**DEVELOPING (1)**

The literature is presented in a disconnected way — either as a chronological list or a summary of unrelated papers. No structure or flow is evident.

PROFICIENT (2)

The literature is partly organized around key themes or periods. Some grouping or flow is attempted but not consistently followed.

EXEMPLARY (2.5)

The literature is clearly and logically grouped into meaningful themes or categories (e.g., approaches, trends, technologies), making it easy to follow the evolution of thought or findings in the field.

Parameter: 4. Identification of Gaps (How well the student identifies what's missing in existing work)**DEVELOPING (1)**

The student fails to articulate what is missing in the existing research, or the gap is too generic or vague to be actionable.

PROFICIENT (2)

A gap is identified with some reference to existing work, though it may not be very specific or well-supported.

EXEMPLARY (2.5)

The student clearly identifies a specific, researchable gap that is justified through the literature review. The gap is connected to future work or objectives of the current study.

MILESTONE: 2. PROBLEM DEFINITION**Marks: 15****Duration: Week 3 and 4****Parameter: 1. Clarity of Problem Statement** (How well the problem is framed and articulated)**DEVELOPING (1)**

The problem is vaguely defined, lacks focus, or is not clearly stated. Key elements such as what is being addressed, and why, are missing or ambiguous.

PROFICIENT (2)

The problem is generally well defined with some clarity on the goal and direction. Some aspects may still be underdeveloped or slightly broad.

EXEMPLARY (3)

The problem is precisely stated, focused, and unambiguous. It clearly communicates what is being addressed, why it matters, and its boundaries.

Parameter: 2. Alignment with Literature (How well the problem connects to existing research)**DEVELOPING (1)**

The problem is disconnected from the literature review or contradicts existing findings. No linkage is provided.

PROFICIENT (2)

The problem shows partial alignment with themes and findings from the literature, though the connection may lack depth.

EXEMPLARY (3)

The problem is logically and strongly derived from the literature review, demonstrating consistency with established research or identified gaps.

MILESTONE: 2. PROBLEM DEFINITION**Marks: 15****Duration:** Week 3 and 4**Parameter: 3. Gap Justification** (How well the student identifies what is missing in existing work)**DEVELOPING (1)**

The gap is either missing, superficially stated, or unrelated to the actual review conducted.

PROFICIENT (2)

The gap is evident and reasonably justified but lacks sharpness or precision in presentation.

EXEMPLARY (3)

The gap is specific, researchable, and directly arises from critical review of literature. The justification is clear and well supported.

Parameter: 4. Feasibility of the Problem (Can the problem be realistically addressed within the time and resource constraints?)**DEVELOPING (1)**

The problem is unrealistic in scope, tools required, or complexity. It is not achievable within the course timeline.

PROFICIENT (2)

The problem is mostly feasible but may have minor scope or complexity issues that could hinder progress.

EXEMPLARY (3)

The problem is well scoped, relevant, and clearly feasible within the 13-week timeline and available resources.

MILESTONE: 2. PROBLEM DEFINITION

Marks: 15

Duration: Week 3 and 4

Parameter: 5. Constraints of the Solution (Have factors like ethics, cost, environment, safety, standards, sustainability been considered?)

DEVELOPING (1)

Constraints are not mentioned or only vaguely acknowledged. No integration into problem framing is evident.

PROFICIENT (2)

Some relevant constraints (e.g., cost or ethics) are identified, but their influence on the problem or approach is not clearly discussed.

EXEMPLARY (3)

Multiple constraints are thoughtfully considered and integrated into the framing of the problem. Trade-offs or priorities are acknowledged.

MILESTONE: 3. METHODOLOGY, EXPERIMENTAL SETUP & RESULTS**Marks: 15****Duration: Week 5 and 8****Parameter: 1. Methodological Planning and Alignment** (How well the method matches the objectives)**DEVELOPING (1)**

The method is poorly defined, lacks structure, or is disconnected from the problem/objectives.

PROFICIENT (2)

The method is generally appropriate and mostly aligned with the problem and objectives. Some gaps may exist in planning.

EXEMPLARY (3)

The method is clearly defined, logically structured, and tightly aligned with the objectives and constraints.

Parameter: 2. Implementation and Technical Soundness (How well the method is executed technically)**DEVELOPING (1)**

Implementation is incomplete or incorrect. Key steps, tools, or components are missing.

PROFICIENT (2)

Implementation is functional with minor technical issues or partial completion.

EXEMPLARY (3)

Implementation is complete, technically robust, and logically sequenced. Demonstrates sound technical proficiency.

MILESTONE: 3. METHODOLOGY, EXPERIMENTAL SETUP & RESULTS

Marks: 15

Duration: Week 5 and 8

Parameter: 3. Resource and Tool Utilization (Use of tools, platforms, or technologies relevant to the domain)**DEVELOPING (1)**

Tools/platforms used are inappropriate, or usage is minimal and ineffective.

PROFICIENT (2)

Tools/platforms are mostly appropriate and used with some clarity.

EXEMPLARY (3)

Tools, libraries, or platforms used are domain-appropriate and skillfully applied. Toolchain is justified and efficient.

Parameter: 4. Design Alternatives and Choice of Approach (Consideration of multiple design flows or strategies)**DEVELOPING (1)**

Only one approach attempted without evaluation of alternatives. No justification provided.

PROFICIENT (2)

One or two alternatives considered with basic rationale for selection.

EXEMPLARY (3)

Multiple viable design flows evaluated; final choice is clearly justified with sound reasoning.

MILESTONE: 3. METHODOLOGY, EXPERIMENTAL SETUP & RESULTS**Marks: 15****Duration:** Week 5 and 8**Parameter: 5. Workflow Clarity and Documentation** (Clarity of the workflow and how well it is documented)**DEVELOPING (1)**

Workflow is ad hoc or undocumented; difficult to follow or reproduce.

PROFICIENT (2)

Workflow is outlined and partially documented; moderate clarity.

EXEMPLARY (3)

Workflow is clearly structured, sequential, and well-documented with diagrams, annotations, or procedural clarity.

MILESTONE: 4. VALIDATION, TESTING & PAPER WRITING

Marks: 20

Duration: Week 9 and 10

Parameter: 1. Validation Strategy and Testing Protocol (How well the solution is tested and validated)

DEVELOPING (2)

Minimal testing conducted; validation lacks structure or relevance. Results not reproducible or incomplete.

PROFICIENT (3)

Basic validation approach applied (e.g., functional, simulation-based, or comparative). Some structure in methodology.

EXEMPLARY (5)

A well-planned, structured validation strategy using domain-appropriate protocols (e.g., simulations, empirical testing, benchmarks). Evidence of repeatability and credibility.

Parameter: 2. Analysis and Interpretation of Results (How well results are analyzed and interpreted)

DEVELOPING (2)

Results are presented without analysis or insight. No linkage to problem statement or objectives.

PROFICIENT (3)

Results are interpreted with some logic and linked to problem objectives. May lack critical insight or depth.

EXEMPLARY (5)

Results are deeply analyzed, with insights on performance, limitations, and connections to objectives. Shows critical thinking and data literacy.

MILESTONE: 4. VALIDATION, TESTING & PAPER WRITING**Marks: 20****Duration:** Week 9 and 10**Parameter: 3. Benchmarking and Performance Comparison** (Comparison with existing methods or references)**DEVELOPING (2)**

No benchmarking or performance comparison attempted. Claims made without reference.

PROFICIENT (3)

Some effort to compare results with existing methods or literature; limited metrics used.

EXEMPLARY (5)

Thorough comparison with existing tools, standards, or solutions using appropriate performance metrics. Demonstrates clear added value or insight.

Parameter: 4. Paper Writing and Submission Readiness (Quality and completeness of paper for publication)**DEVELOPING (2)**

Draft is poorly structured, lacks clarity, or not prepared for submission. Formatting or referencing not followed.

PROFICIENT (3)

Paper is mostly well-structured and readable. Minor formatting or content issues. Submitted to a reasonable venue.

EXEMPLARY (5)

Paper is polished, well-structured, and prepared as per submission guidelines (conference/journal/patent). Submitted to a relevant and reputable platform (e.g., SCOPUS-indexed, IPO).

MILESTONE: 5. END-TERM VIVA & FINAL EVALUATION

Marks: 40

Duration: Week 11 and 13

Parameter: 1. Conceptual Understanding and Response Accuracy (How well the student demonstrates understanding of the domain and responds to questions)

DEVELOPING (3)

Struggles with core concepts; answers are vague, inaccurate, or inconsistent with project.

PROFICIENT (5)

Demonstrates general understanding with reasonably accurate responses to most questions.

EXEMPLARY (8)

Shows clear, deep understanding of concepts and confidently handles complex and context-specific questions with precision.

Parameter: 2. Justification of Technical Decisions (How well the student can justify methodology, tools, and design choices)

DEVELOPING (2)

Unable to clearly explain why specific methods, tools, or strategies were chosen.

PROFICIENT (3)

Offers partial or general reasoning for decisions made in the project.

EXEMPLARY (5)

Provides detailed, logical, and well-reasoned justifications for all key decisions, with alternatives considered where applicable.

MILESTONE: 5. END-TERM VIVA & FINAL EVALUATION**Marks: 40****Duration: Week 11 and 13****Parameter: 3. Communication and Presentation Effectiveness** (Quality of explanation, presentation style, and clarity of expression)**DEVELOPING (3)**

Presentation is disorganized, difficult to follow, or overly dependent on reading slides.

PROFICIENT (5)

Presentation is mostly clear and organized; minor lapses in flow or depth.

EXEMPLARY (8)

Well-structured, engaging, and articulate delivery. Visuals and speech complement each other. Answers demonstrate reflective preparation.

Parameter: 4. Outcome Achieved (Paper/Patent Submission) (Submission of outcome to appropriate platform and its quality)**DEVELOPING (2)**

Outcome not submitted or lacks required structure; insufficient for evaluation.

PROFICIENT (3)

Submitted to an acceptable conference/journal/patent portal with reasonably complete content.

EXEMPLARY (5)

Submitted to a SCOPUS-indexed journal/conference or filed with IPO; content meets publication standards and reflects substantial work.

MILESTONE: 5. END-TERM VIVA & FINAL EVALUATION

Marks: 40

Duration: Week 11 and 13

Parameter: 5. Constraints Addressed in Final Solution (Consideration of ethics, cost, sustainability, safety, and applicable standards in the final work)

DEVELOPING (3)

Constraints ignored or superficially mentioned without meaningful integration.

PROFICIENT (5)

Some constraints identified and loosely addressed in design or implementation.

EXEMPLARY (8)

Multiple real-world constraints (e.g., ethics, sustainability, safety, cost, and applicable standards) are consciously integrated into the final design and discussed during viva.



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